Adam Sheikhdon

Aurora,ON | 647-540-4436 | Adamsheik621@gmail.com | LinkedIn | Portfolio

Education

Western University London, ON

Bachelor of Electrical Engineering Sciences *September 2024 – Present*

- **Relevant Coursework:** Digital Logic, Electronics, Heat Transfer, Circuits & Systems, Magnetic Circuits

Extracurricular

Western Engineering Mars Rover Team London, ON

Electrical Team *September 2024 – Present*

- Member of a 50+ student team designing a Mars rover; developed and optimized the power distribution network to ensure reliable energy delivery to all subsystems.
- Designed and tested power distribution PCBs in Altium Designer, coordinating with the electrical subteam to meet subsystem power requirements.

Projects

24V to 12V Buck Converter | *Altium Designer*

- Designed and built a regulated buck converter PCB using Altium Designer for the Mars Rover, providing a stable 12V output to power rover subsystems.
- Applied power electronics principles to select appropriate components, ensuring stable regulation and efficiency under the rover's load requirements.

24V to 5V Buck Converter | *Altium Designer*

- Designed and routed a custom 24V to 5V buck converter PCB in Altium Designer to safely power sensitive 5V logic and sensors on the rover.
- Optimized footprint selection and component sizing for the switching regulator's input capacitance, boost circuitry, and output filtering to ensure reliable power distribution across the board.

Dual-Input Power Distribution Board | *Altium Designer*

- Designed a custom PCB in Altium Designer combining two XT90 inputs into a single XT90 output rated for 60A, with an integrated circuit breaker for overcurrent protection and a relay for remote switching.
- Sized PCB traces and protection components to meet the 60A rating while maintaining thermal and electrical safety margins.

Work Experience

Western University London, ON

Lab Assistant, Work-Study *October 2026 – Present*

- Contribute to the development of an autonomous RC car that uses AI-based perception to navigate its environment, supporting embedded systems and sensor integration.
- Support the lab supervisor with hands-on hardware work, including wiring and cable routing, soldering, and assembling and testing prototypes.

Cineplex Aurora, ON

Cast Member *November 2023 – August 2026*

- Assisted 500+ daily guests with ticket purchases, concessions, and seating inquiries to ensure a positive moviegoing experience.
- Recognized as Employee of the Month for exceptional performance and dedication to guest service.

Skills

Electrical Engineering Tools: Altium Designer, MicroCap, Quartus Prime, OnShape
Hardware Prototyping Techniques: Soldering, Wiring, Multimeters & Oscilloscopes, Breadboard Circuits
Programming Languages: Java, Python, MATLAB
Software: Microsoft Word, PowerPoint, Excel